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Advancing Hydraulic Fracture Monitoring: Elastic Full Waveform Inversion With Moment Tensor Implementation Using Das And/or Geophones

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ABSTRACT

Hydraulic fracture monitoring (HFM) is critical to optimizing reservoir stimulation and maximizing hydrocarbon recovery. Traditional microseismic methods rely primarily on kinematic data, involving manual time-picking and waveform editing, which limits robustness under low-to-moderate signal-to-noise ratios (SNR). This paper presents a novel methodology integrating elastic full waveform inversion (eFWI) with moment tensor inversion (MTI), leveraging fused Distributed Acoustic Sensing (DAS) and geophone data. The hybrid approach harnesses the full seismic wavefield, eliminating the need for manual picking and enabling high-resolution source mechanism characterization. We develop a 3D elastic wave propagation solver incorporating moment tensor sources and apply an adjoint-based inversion framework for simultaneous velocity, anisotropy, event location, and moment tensor estimation. Sensitivity and resolution matrix analyses guide sensor deployment and assess parameter recoverability. Validation with synthetic and field data demonstrates strong waveform reconstruction and accurate moment tensor recovery. The method differentiates tensile, shear, and hybrid fracture mechanisms, providing critical insights for optimizing hydraulic fracturing design and reducing seismic risk. Our results highlight the transformative potential of eFWI-MTI fusion in advancing HFM capabilities and reservoir management.

INTRODUCTION

Hydraulic Fracture Monitoring (HFM) is important for improving reservoir stimulation and increasing hydrocarbon production (Le Calvez, et al., 2017). Traditional microseismic methods mainly use kinematic data like event arrival times and often miss valuable information found in the full seismic waveforms. These methods also depend on manual time-picking, which can be difficult and less reliable when the signal-to-noise ratio (SNR) is low to moderate (Maxwell, 2014).

Distributed Acoustic Sensing (DAS) has become more popular in seismic monitoring because it can turn fiber optic cables into dense arrays of strain sensors (Hartog, 2017; Li, et al., 2021). However, traditional microseismic processing methods have a hard time fully using the unique data provided by DAS (Staněk, et al., 2022). This study introduces a new approach that combines Elastic Full Waveform Inversion (eFWI)

with Moment Tensor (MT) inversion, using both geophone and DAS data to better understand hydraulic fracturing.

Moment Tensor Inversion (MTI) is a useful tool for understanding the source of seismic events. It is widely used in earthquake studies and increasingly in hydraulic fracture monitoring. MTI breaks down seismic signals to reveal the physical processes causing the events (Aki & Richard, 2002; Vavryčuk, 2015). In hydraulic fracturing, MTI helps optimize production and reduce the risk of unwanted seismic activity (Tape & Tape, 2015).

Moment tensor inversion has developed over many years, from early analytical methods for large earthquakes to modern computational techniques. Full waveform inversion (FWI), which uses the whole seismic waveform rather than just the first arrivals, has improved the accuracy of source characterization (Virieux & Operto, 2009; Pratt, 1999). Recently, MTI has been applied to smaller, induced seismic events from hydraulic fracturing and geothermal energy. Combined with real-time data and advanced algorithms, this allows quick determination of source mechanisms, providing better insights into subsurface stress and fracture behavior (Aki & Richard, 2002).

ADVANCEMENTS IN eFWI FOR HFM

Recent developments have made the application of Elastic Full Waveform Inversion (eFWI) in Hydraulic Fracture Monitoring (HFM) more effective but also more complex.

Utilization of DAS in HFM

Distributed Acoustic Sensing (DAS) has gained wide use due to its versatility and operational advantages, especially in time-lapse monitoring. DAS records strain or strain-rate along a fiber optic cable over a gauge length (Ma, et al., 2023). This means that the DAS response depends on the well trajectory and gauge length (Li, et al., 2021), which need to be carefully considered in modeling and inversion. DAS fibers can be installed in two main ways: permanently behind the casing, offering better coupling with the formation and higher signal-to-noise ratio (SNR), or temporarily inside the tubing for shorter-term monitoring. Additionally, fiber optic cables can be deployed on the seafloor for marine seismic acquisition (Rashid, et al., 2025). These installation methods impact the quality and type of data collected (Hartog, 2017).

DAS transforms fiber optic cables into dense arrays of strain sensors, enabling near real-time monitoring of hydraulic fracturing. This provides detailed insights into fracture growth and reservoir behavior (Wang, et al., 2022). For example, the Canadian Dip-in DAS (CanDiD-1) project demonstrated the use of retrievable optical fibers to monitor hydraulic fractures effectively, capturing both microseismic events and low-frequency signals (Wang, et al., 2022).

eFWI in Anisotropic Media

Subsurface formations often exhibit anisotropic properties, which can significantly affect seismic wave propagation. FWI methods have been adapted to account for anisotropy, improving the accuracy of subsurface imaging and characterization of complex geological features. Recent studies have developed practical frameworks for elastic FWI in anisotropic media, focusing on resolving parameter trade-offs and stabilizing inversion processes.

Advancements in Moment Tensor Inversion

Incorporating anisotropy into Moment Tensor Inversion (MTI) enables more precise determination of microseismic event mechanisms. This is crucial for understanding fracture orientations and local stress fields (Li, et al., 2020). The development of efficient modeling tools that simulate wave propagation in anisotropic media has further improved the accuracy of MTI and source mechanism analysis.

MOMENT TENSOR INVERSION AND ITS APPLICATION IN HYDRAULIC FRACTURE MONITORING

Moment Tensor Inversion (MTI) is a valuable technique for characterizing the source mechanisms of seismic events. Widely applied in both earthquake seismology and induced seismicity studies, MTI helps decompose recorded wavefields into fundamental source types, offering insight into the physical processes governing fracture generation and propagation. In the context of Hydraulic Fracture Monitoring (HFM), MTI contributes to improved interpretation of fracture behavior, aiding in the optimization of stimulation strategies and mitigation of seismic hazards.

The moment tensor represents the spatial distribution of forces acting at the seismic source. It is a symmetric 3×3 matrix that encodes information about the direction, intensity, and type of stress within the medium. The general form of the moment tensor matrix, M is:

$$M = \begin{bmatrix} M_{xx} & M_{xy} & M_{xz} \\ M_{xy} & M_{yy} & M_{yz} \\ M_{xz} & M_{yz} & M_{zz} \end{bmatrix} \quad (1)$$

Where M_{xx} , M_{yy} , M_{zz} represent the normal (diagonal) components, which relate to the volumetric and deviatoric stress.

M_{xy} , M_{xz} , M_{yz} are the off-diagonal components representing the shear stress.

The eigenvalues of the moment tensor matrix can help determine the nature of the seismic event. In particular, the matrix can be used to identify different types of source mechanisms based on the relative magnitudes of its components (Aki & Richard, 2002).

Interpretation of Moment Tensor Components

The moment tensor matrix provides a detailed description of the seismic source. There are several key types of source mechanisms that can be identified based on the moment tensor components:

Isotropic (Explosion) Source

Characterized by equal, positive diagonal components and zero off-diagonal terms, this source type corresponds to a volumetric expansion, commonly associated with explosive or tensile fracture events. Such mechanisms indicate effective fracture opening driven by fluid injection.

The moment tensor matrix for such a source is diagonal with equal diagonal elements:

$$M_{ISO(EXP)} = \begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & M_{xx} & 0 \\ 0 & 0 & M_{xx} \end{bmatrix} \quad (2)$$

M_{xx} here is positive. This moment tensor matrix corresponds to a point source where the energy is radiated equally in all directions.

Double-Couple (DC) or Shear Faulting Source

Double-couple mechanism occurs when the seismic event is dominated by shear stress along a fault plane. The moment tensor matrix for a double-couple source, which is typical for earthquakes, takes the following form described by non-zero off-diagonal components with no volumetric change:

$$M_{DC} = \begin{bmatrix} M_{xx} & M_{xy} & M_{xz} \\ M_{xy} & -M_{xx} & 0 \\ M_{xz} & 0 & 0 \end{bmatrix} \quad (3)$$

This matrix represents a shear fault where the normal stress on the fault is opposed by shear forces, which cause displacement along the fault plane (Willacy, et al., 2019).

Tensile (Crack Opening) Source

A purely tensile source shows non-zero diagonal terms corresponding to extension, with minimal or zero shear components. This mechanism is of particular interest in stimulation operations aiming to maximize fracture aperture. A tensile source, can be represented as:

$$M_{Tensile} = \begin{bmatrix} M_{xx} & 0 & 0 \\ 0 & M_{yy} & 0 \\ 0 & 0 & M_{zz} \end{bmatrix} \quad (4)$$

Combined or Hybrid Source Mechanism

In many real-world seismic events, especially in induced seismicity from hydraulic fracturing, the moment tensor matrix is a combination of isotropic, deviatoric, and tensile components. These complex mechanisms represent a combination of shear faulting and tensile fracture, often occurring simultaneously during hydraulic fracturing operations (Zoback & Harjes, 1997).

Understanding these components is crucial for interpreting whether a fracture is opening, sliding, or compacting, and for evaluating the effectiveness and safety of stimulation.

Applications in Hydraulic Fracturing

During hydraulic fracturing, high-pressure fluid injection induces fractures in the rock. These processes generate seismic signals whose analysis through MTI can provide:

Fracture Mode Classification

Differentiating between tensile and shear events is essential. While tensile opening is typically desirable for reservoir stimulation, shear events may indicate slip on natural faults and potential seismic risk.

Stress Field Characterization

The orientation and magnitude of moment tensor components help infer the in-situ stress regime and fracture orientations, enabling better planning of stage intervals and perforation strategies.

Event Magnitude and Source Size

The strength and spatial distribution of moment tensors relate to the energy release and rupture dimensions, offering insight into the stimulated rock volume.

Focal Mechanism Categories and Fracturing Implications

Below table summarizes different categories of focal mechanisms, associated moment tensor components and it's relevance in regards to hydraulic fracturing for production optimization and enhancement.

Table 1—Different categories of focal mechanisms, associated moment tensor components and it's relevance in regards to hydraulic fracturing for production optimization and enhancement

Category	Physical Interpretation	Representative MT	Fracturing Relevance
Tensile Opening	Fluid-driven crack opens perpendicular to minimum stress	$\begin{bmatrix} + & 0 & 0 \\ 0 & + & 0 \\ 0 & 0 & - \end{bmatrix}$	Highly desirable – effective fracture creation
Shear Slip (DC)	Slip on pre-existing fracture/fault	$\begin{bmatrix} 0 & + & 0 \\ + & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	Adds to SRV, but may indicate fault reactivation
Hybrid (Tensile + Shear)	Mixed tensile and shear behavior in complex rock fabric	$\begin{bmatrix} + & + & 0 \\ + & - & 0 \\ 0 & 0 & - \end{bmatrix}$	Desirable – complex, connected fracture networks
Collapse (–ISO)	Volume contraction or compaction	$\begin{bmatrix} - & 0 & 0 \\ 0 & - & 0 \\ 0 & 0 & - \end{bmatrix}$	Undesirable – may indicate ineffective stimulation

Moment tensor inversion provides a deeper understanding of the mechanical processes involved in hydraulic fracturing. By distinguishing between shear and tensile failure, and assessing the stress orientation and rupture dynamics, MTI contributes significantly to reservoir evaluation and risk mitigation. It enables operators to refine fracture designs, enhance stimulated rock volume, and reduce the potential for induced seismicity.

CHALLENGES AND PROPOSED SOLUTIONS FOR eFWI IN HFM WITH DAS/GEOPHONE RECORDINGS

The integration of Distributed Acoustic Sensing (DAS) with conventional geophones has emerged as a promising strategy in seismic monitoring (Li, et al., 2023), offering enhanced spatial resolution and continuous coverage. However, this hybrid setup introduces a set of technical and computational challenges that must be addressed for effective application in Full Waveform Inversion (FWI) for Hydraulic Fracture Monitoring (HFM).

Data Fusion Challenges

DAS vs. Geophones: DAS and geophones have inherently different measurement principles. DAS records axial strain or strain rate along the optical fiber, while geophones measure particle velocity. The conversion of DAS measurements into geophone-equivalent data (or vice versa) often introduces transformation errors that can degrade inversion accuracy. These errors arise due to differences in sensor coupling, orientation, gauge length, and frequency response, complicating joint inversion workflows. Particularly, inaccuracies in strain-to-velocity transformation impact the amplitude fidelity, leading to biased moment tensor solutions and incorrect magnitude estimates (Staněk, et al., 2022).

To overcome these issues, a sensor-domain-aware approach is being adopted. Rather than converting between data domains, forward modeling and inversion processes are performed directly on the native data types—strain for DAS and velocity for geophones. This approach reduces transformation artifacts and ensures that the inversion leverages the true nature of each sensor type.

Limitations of Acoustic-Only Modeling

Early FWI applications in hydraulic fracturing often relied on acoustic wavefield simulations, which model only pressure (P) waves and ignore shear (S) waves. This simplification limits the ability to resolve elastic properties and identify complex fracture mechanisms. Recent work has expanded the use of elastic forward modeling, which simulates both stress and displacement fields and captures full P- and S-wave behavior, essential for realistic source mechanism interpretation (Wang & Alkhalifah, 2018; Willacy, et al., 2019).

Elastic modeling is particularly important when interpreting DAS data, as the axial strain response is sensitive to both P- and S-waves, and their interactions with fracture geometry. Incorporating elastic effects enables more accurate representation of the wavefield and better resolution of anisotropic features and fracture-induced heterogeneities.

Sparse Acquisition and Resolution Degradation

eFWI typically assumes dense and uniform receiver coverage. However, in field deployments—especially in unconventional reservoirs—acquisition may be sparse due to economic or logistical constraints. Sparse arrays can result in spatial aliasing, reduced inversion stability, and poor resolution of small-scale features.

To mitigate these issues, sparsity-promoting regularization methods have been incorporated into the inversion process. These techniques—such as L1-norm constraints and total variation minimization—enhance the robustness of inversion in underdetermined conditions. Shekar and Sethi (2019) demonstrated the effectiveness of such methods in improving resolution in sparse acquisition settings, and similar strategies have been adopted in the present study.

Computational Considerations

The real-time processing of large-scale DAS and geophone datasets remains computationally intensive. Forward modeling and adjoint computations in elastic media, especially with high spatial resolution, require significant memory and processing power. Optimization strategies such as domain decomposition, GPU acceleration, and adaptive meshing are being employed to reduce computational overhead and facilitate near-real-time analysis in operational environments. The work on this front is still under progress.

THEORETICAL FRAMEWORK FOR SEISMIC DATA ACQUISITION: DISPLACEMENT AND STRAIN SENSORS

Geophone Displacement

Geophones measure ground motion in terms of displacement or particle velocity. The displacement field u_n resulting from a seismic source can be described using the moment tensor formulation and Green's function formalism, as outlined by [Aki and Richards \(2002\)](#). The general form is:

$$u_n = G_{np,q} * M_{pq} = \iint \frac{\partial G_{np}(x, t, \xi, \tau)}{\partial \xi_q} M_{pq}(\xi, \tau) d\xi \quad (5)$$

This leads to the analytical form of the displacement field from a point source at distance r :

$$\begin{aligned} u_n(t) = & \left(\frac{15\gamma_n\gamma_p\gamma_q - 3\gamma_n\delta_{pq} - 3\gamma_p\delta_{nq} - 3\gamma_q\delta_{np}}{4\pi\rho} \right) \frac{1}{r^4} \int_{r/\alpha}^{r/\beta} \tau M_{pq}(t - \tau) d\tau \\ & + \left(\frac{6\gamma_n\gamma_p\gamma_q - \gamma_n\delta_{pq} - \gamma_p\delta_{nq} - \gamma_q\delta_{np}}{4\pi\rho\alpha^2} \right) \frac{1}{r^2} M_{pq}\left(t - \frac{r}{\alpha}\right) - \left(\frac{6\gamma_n\gamma_p\gamma_q - \gamma_n\delta_{pq} - \gamma_p\delta_{nq} - 2\gamma_q\delta_{np}}{4\pi\rho\beta^2} \right) \frac{1}{r^2} M_{pq}\left(t - \frac{r}{\beta}\right) \\ & + \frac{\gamma_n\gamma_p\gamma_q}{4\pi\rho\alpha^3} \frac{1}{r} M_{pq}\left(t - \frac{r}{\alpha}\right) - \frac{(\gamma_n\gamma_p - \delta_{np})\gamma_q}{4\pi\rho\beta^3} \frac{1}{r} M_{pq}\left(t - \frac{r}{\beta}\right) \end{aligned} \quad (6)$$

Where:

- u_n : n-th component of the Displacement
- $G_{np,q}$: Partial derivative of Green's function with respect to source location,
- M_{pq} : Moment tensor components,
- γ : Direction cosines from source to receiver,
- δ_{ij} : Kronecker delta,
- ρ : Density of the medium,
- α : P-wave velocity,
- β : S-wave velocity,
- r : Source–receiver distance,
- t : Time variable.

The contributions to the wavefield sensitivity are divided based on the scaling with distance r and the type of waves (P- wave α , S-wave β). This formulation includes **near-field** ($\sim 1/r^4$), **intermediate-field** ($\sim 1/r^2$), and **far-field** ($\sim 1/r$) terms ([Aki & Richard, 2002](#)), corresponding to contributions from different wave types and propagation zones.

DAS strain projected along fiber

Unlike geophones, Distributed Acoustic Sensing (DAS) systems measure **strain** or **strain-rate** along the direction of a fiber-optic cable. This distinction introduces both opportunities and challenges in data interpretation and inversion.

The full strain tensor ϵ , which describes deformation in an elastic medium, is defined in terms of the spatial derivatives of displacement components as:

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (7)$$

Here, u_i and u_j are displacement components in the i -th and j -th directions, and x_j and x_i are spatial coordinates.

DAS systems, however, do not measure the full strain tensor. Instead, they record the strain projected along the optical fiber direction (Wang, et al., 2022; Vera Rodriguez & Wuestefeld, 2020), characterized by a unit tangent vector t . The DAS-recorded strain S_{DAS} is obtained through a projection of the strain tensor along the fiber axis. The measured strain S_{DAS} is:

$$S_{DAS} = t^T \epsilon t = t_i t_j \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (8)$$

Where t_i and t_j are the components of the fiber's tangent vector in the i -th and j -th directions, respectively. This projection highlights several key characteristics:

Directional Sensitivity

DAS is only sensitive to deformation along the fiber axis. This makes it highly responsive to axial strain but insensitive to transverse components.

Scalar Measurement

Unlike geophones that capture vector quantities (e.g., three-component velocity), DAS strain is a scalar, influenced by the fiber orientation relative to the wavefield.

Information Loss

A single fiber orientation cannot capture the full strain tensor, potentially limiting resolution unless multiple orientations are available or fused with geophone data.

These factors necessitate customized modeling and inversion strategies when DAS is integrated into a broader seismic monitoring system (Staněk, et al., 2022; Yust, et al., 2023).

MOMENT TENSOR FORMULATION: RESOLUTION MATRIX, SENSITIVITY ANALYSIS, AND INVERSION

The moment tensor formulation establishes a linear relationship between the observed seismic wavefield and the source mechanism (Virieux & Operto, 2009). In its compact form, this relationship is expressed as:

$$\mathbf{u} = \mathbf{G}\mathbf{M} \quad (9)$$

where $\mathbf{u}$ is the observed displacement vector, $\mathbf{G}$ is the Green's function matrix (Pasyanos, et al., 2023) that models wave propagation through the Earth, and $\mathbf{M}$ is the moment tensor vector representing the seismic source mechanism (Aki & Richard, 2002; Menke, 2012).

To estimate $\mathbf{M}$ from recorded data, one typically solves an inverse problem using least-squares or generalized inversion methods. The estimated moment tensor, $\hat{\mathbf{M}}$, is given by:

$$\hat{\mathbf{M}} = \mathbf{G}^\dagger \mathbf{u} \quad (10)$$

where $G^\dagger$ denotes the Moore-Penrose pseudoinverse of the Green's function matrix (Weaver, et al., 2008). The fidelity of this inversion is evaluated using the resolution matrix $\mathbf{R}$, defined as:

$$\mathbf{R} = \mathbf{G}^\dagger \mathbf{G} \quad (11)$$

An ideal resolution matrix approximates the identity matrix, implying that the inversion accurately retrieves each model parameter independently. In this context, the diagonal elements of $\mathbf{R}$ indicate the degree to which individual moment tensor components are recovered, while the off-diagonal elements quantify parameter coupling or trade-offs.

A strongly diagonal resolution matrix indicates good data coverage, minimal parameter cross-talk, and a well-posed inversion. Conversely, large off-diagonal elements signal ambiguity and correlation among components, often resulting from limitations in receiver geometry, low signal-to-noise ratio, or poor conditioning of the forward operator.

The sensitivity matrix, derived from spatial derivatives of the Green's function, provides further insight into how different moment tensor components influence the observed wavefield. For geophones, this sensitivity is based on displacement or velocity fields, while for DAS it is computed based on strain recorded along the fiber. Specifically, the Green's function derivative $G_{np,q}$ characterizes the response at receiver component n due to a point source force in direction q , acting at location p (Aki & Richards, 2002).

Equation (6), which accounts for near-field ($1/r^4$), intermediate-field ($1/r^2$), and far-field ($1/r$) contributions, is central to constructing these sensitivity matrices and evaluating the potential recoverability of moment tensor elements. For DAS, a similar formulation applies but in the strain domain, requiring careful projection of the elastic wavefield onto the fiber direction.

In practical applications, visualizing the resolution matrix helps identify which components of the moment tensor are reliably constrained. For instance, in borehole monitoring geometries with limited azimuthal coverage—such as a single vertical observation well—components like M_{zz} , which represent vertical tensile motion, may be poorly resolved due to the lack of horizontal aperture and directional sensitivity. In contrast, configurations involving both vertical and horizontal observation wells (e.g., one vertical and one horizontal, or two orthogonal horizontal wells) provide improved azimuthal coverage and directional sensitivity, enhancing the resolution of all moment tensor components, including M_{zz} . By analyzing the resolution matrix before and after inversion, one can diagnose inversion stability, improve acquisition design, and apply regularization to stabilize ill-posed systems.

Thus, the resolution matrix is not only a post-inversion diagnostic but also a proactive tool in survey planning and inversion strategy design.

WORKFLOW: ELASTIC FWI WITH MOMENT TENSOR INVERSION

This study presents a modular, physics-driven workflow for Elastic Full-Waveform Inversion (eFWI) integrated with Moment Tensor Inversion (MTI) tailored for hydraulic fracture monitoring (HFM). Figure 1 illustrates the complete workflow, highlighting the interplay between forward modeling, inversion, validation, and iterative model updates.

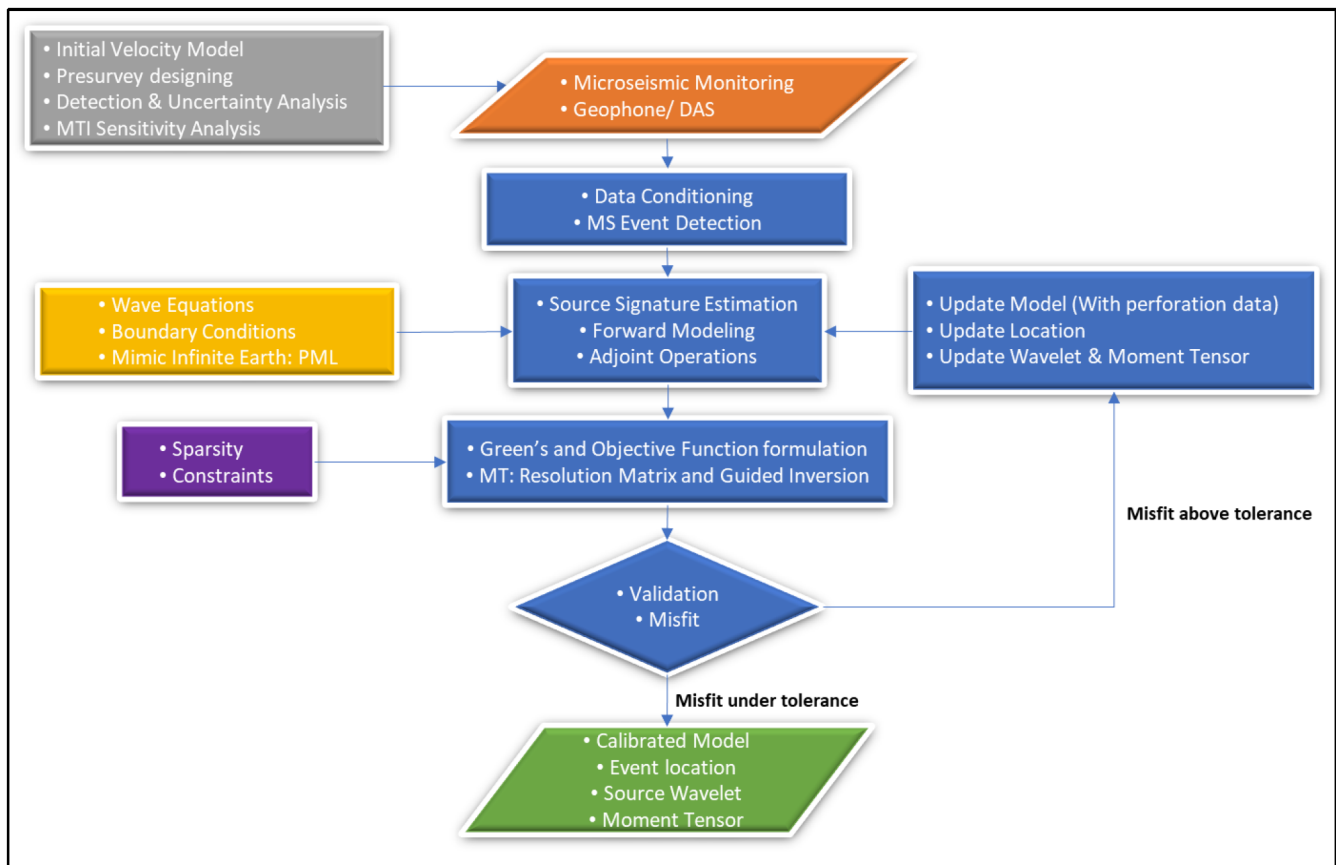


Figure 1—Workflow for microseismic monitoring incorporating Moment Tensor Inversion

The workflow is compatible with both borehole geophones and Distributed Acoustic Sensing (DAS) systems and comprises the following key components:

1. Initial Velocity Model and Pre-Survey Design

A robust pre-survey design is critical for the success of eFWI. This stage focuses on building an initial elastic velocity model using available geophysical and geological information. A sensitivity analysis is then performed to evaluate detection thresholds, inversion uncertainties, and the resolvability of individual moment tensor components.

Trial moment tensor configurations are forward-modeled to assess the directional sensitivity of the planned sensor array. This informs optimal sensor placement and ensures that both DAS and geophone data contribute effectively to source characterization.

2. Microseismic Monitoring and Event Detection

Microseismic data are collected using borehole-deployed geophones or DAS fibers. The raw recordings undergo signal conditioning, denoising, and automatic event detection using algorithms such as Short-Time Average over Long-Time Average (STA/LTA) or machine learning-based classifiers. This process outputs a high-confidence set of candidate events suitable for further analysis.

3. Elastic Forward Modeling

Elastic wave propagation is modeled using a custom 3D finite-difference time-domain (FDTD) solver incorporating full moment tensor (MT) source representations. This modeling simulates realistic wavefields—capturing both P- and S-wave modes—through anisotropic, heterogeneous subsurface models.

Perfectly Matched Layers (PMLs) are applied at the boundaries to simulate an infinite medium. Perforation and well completion data are incorporated into the model, improving the accuracy of

simulated source signatures. Event locations are determined using modified FWI approaches (Shekar & Sethi, 2019) and are further validated using conventional arrival-time-based location methods.

4. **Moment Tensor Inversion and Resolution Matrix Analysis**

A core part of the workflow is the moment tensor inversion, where the Green's function matrix links the observed wavefield to the source mechanism. The inversion is stabilized using the Moore-Penrose pseudoinverse. A resolution matrix is computed to assess the recoverability of each tensor component and identify potential trade-offs. High diagonal dominance in the resolution matrix implies strong invertibility, while off-diagonal terms indicate parameter coupling or limited sensitivity due to acquisition geometry (Menke, 2012; Aki & Richard, 2002).

5. **Validation and Misfit Evaluation**

Inversion outputs are validated through misfit calculations comparing synthetic and observed traces across multiple components and receiver types. Misfit metrics (e.g., L2 norm, normalized cross-correlation) guide further iteration. Model updates continue until a satisfactory match is achieved.

The final calibrated model resulting from this workflow includes accurately located microseismic events, estimated source wavelets, and complete moment tensor solutions. These outputs provide critical insights into the fracture mechanisms and stress orientations within the subsurface and allow for more precise estimation of the stimulated reservoir volume (SRV). Such detailed characterizations are essential for evaluating the effectiveness of hydraulic fracturing operations and optimizing production strategies (Feng, et al., 2019).

A few key advantages distinguish this workflow from conventional approaches. Firstly, it eliminates the need for manual time-picking by employing a fully automated framework that leverages full waveform data. Secondly, it integrates hybrid sensing technologies, combining the dense spatial sampling of Distributed Acoustic Sensing (DAS) with the multi-component capabilities of borehole geophones to maximize observational coverage and fidelity. Additionally, the inversion process is physically informed—guided by accurate Green's functions, moment tensor sensitivity analysis, and well-defined constraints—to ensure robust and interpretable results. Finally, the workflow is both scalable and iterative, allowing for real-time updates and efficient batch processing, making it suitable for dynamic reservoir monitoring and operational decision-making.

RESULTS AND OBSERVATIONS

The implementation of the Elastic Full-Waveform Inversion (eFWI) framework, integrated with Moment Tensor Inversion (MTI), demonstrated significant advancements in microseismic source characterization using both synthetic and field datasets. This approach directly addresses longstanding challenges in hydraulic fracture monitoring, particularly those associated with manual time-picking and reliance on simplified kinematic models. By leveraging waveform data in their native measurement domains—displacement for geophones and strain for DAS—the framework enables fully automated, high-fidelity inversion of microseismic events with improved spatial resolution and interpretability.

Synthetic Data Analysis

To validate the robustness and accuracy of the MTI framework, synthetic datasets were generated using predefined moment tensor sources embedded in realistic elastic velocity models. The goal was to assess the resolution of individual moment tensor components under idealized acquisition geometries for both geophones and DAS.

Figure 2 illustrates the results for geophone-based recordings. The left panel presents the acquisition geometry designed to ensure wide azimuthal coverage and multi-component wavefield recording. The center panel shows the computed resolution matrix, where strong diagonal dominance indicates excellent

component separation and minimal parameter trade-off. The right panel compares the true (input) and inverted moment tensors, confirming accurate recovery of all resolvable source components.

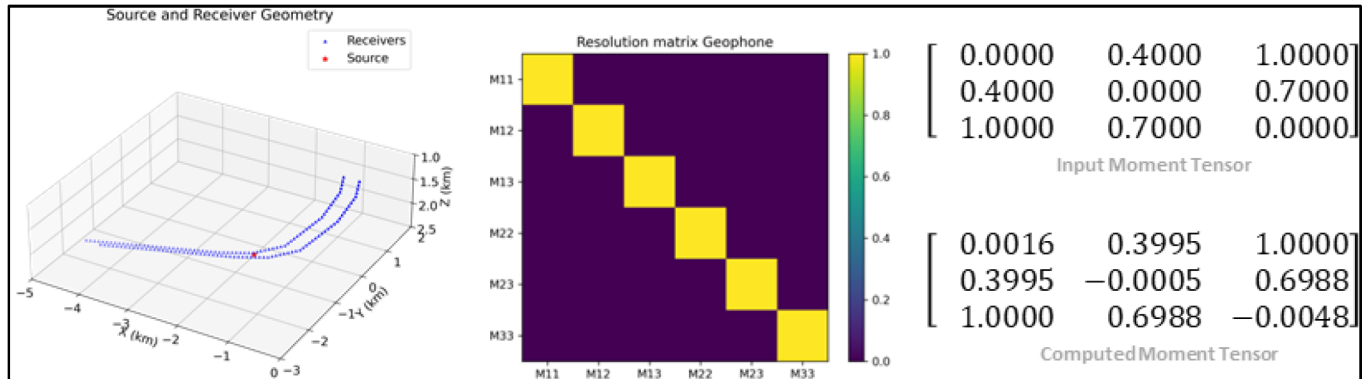


Figure 2—Synthetic Moment Tensor Inversion Results – Geophone-Based Acquisition
 (Left) Acquisition geometry for geophone receivers ensuring multi-azimuthal coverage. (Middle) Computed resolution matrix showing strong diagonals and minimal off-diagonal leakage. (Right) Comparison between input and inverted moment tensor components, confirming high inversion fidelity

Similarly, **Figure 3** presents synthetic test results for DAS-based recordings. The acquisition layout (left panel) simulates a fiber-optic cable aligned along a wellbore, capturing strain measurements along its length. While the resolution matrix (middle panel) shows slight degradation in off-diagonal elements due to DAS's directional sensitivity, the principal tensor components remain well resolved. The right panel again shows strong agreement between true and inverted moment tensors, validating the method across sensor modalities.

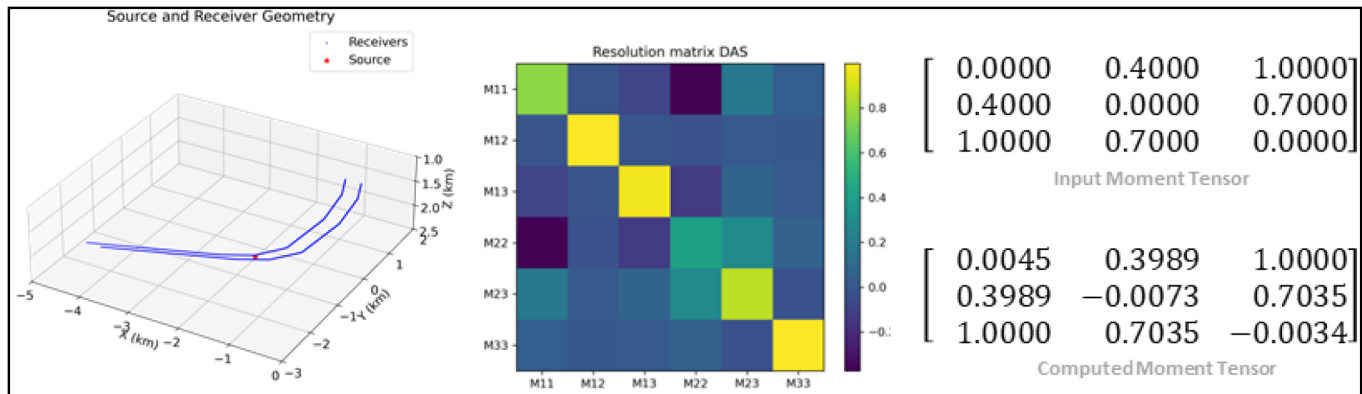


Figure 3—Synthetic Moment Tensor Inversion Results – DAS-Based Acquisition
 (Left) Simulated fiber layout for DAS strain measurements. (Middle) Resolution matrix indicating directional sensitivity, with significant off-diagonal leakages and five of six components decently resolved. (Right) Accurate reconstruction of the input moment tensor through inversion, confirming DAS viability for MTI.

Field Data Analysis

The eFWI-MTI framework was further tested on field data from a vertical borehole microseismic monitoring program that employed both geophones and DAS installations. Two microseismic events—designated Event-1 and Event-2—were selected for detailed analysis based on waveform clarity, spatial coverage, and signal-to-noise ratio.

Figure 4 presents the 3D visualization of the acquisition geometry and event locations. The green cylinders represent geophone receiver positions along the monitoring well, while colored spheres indicate perforation clusters across different hydraulic fracturing stages. Event-1 (blue) and Event-2 (yellow) are highlighted within the stimulated zone.

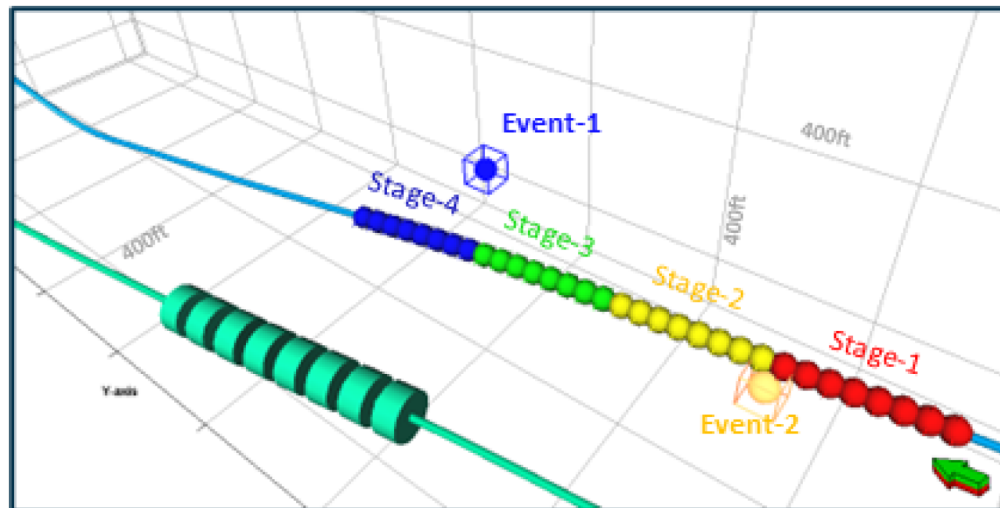


Figure 4—Field Acquisition Geometry and Microseismic Event Locations
Visualization of geophone receiver positions (green cylinders), perforation clusters (colored spheres), and microseismic event locations—Event-1 (blue sphere) and Event-2 (yellow sphere)—within the hydraulic stimulation volume.

The recorded waveforms for Event-1 and Event-2 are shown in Figure 5 and Figure 6, respectively. Each trace includes the three orthogonal components—P, Sh, and Sv—depicted in blue, red, and green. On the left side of each figure, circular hodogram plots illustrate the polarization angles for each arrival. Both events exhibit well-defined P, Sh, and Sv phases with no ambiguity in arrival identification. However, Event-1 displays broader polarization vectors across the hodograms, which enhances directionality and provides better sensitivity for moment tensor inversion. In contrast, although Event-2 also exhibits clear arrivals, its polarization vectors are more constrained, limiting directional coverage and reducing the robustness of moment tensor resolution for certain components.

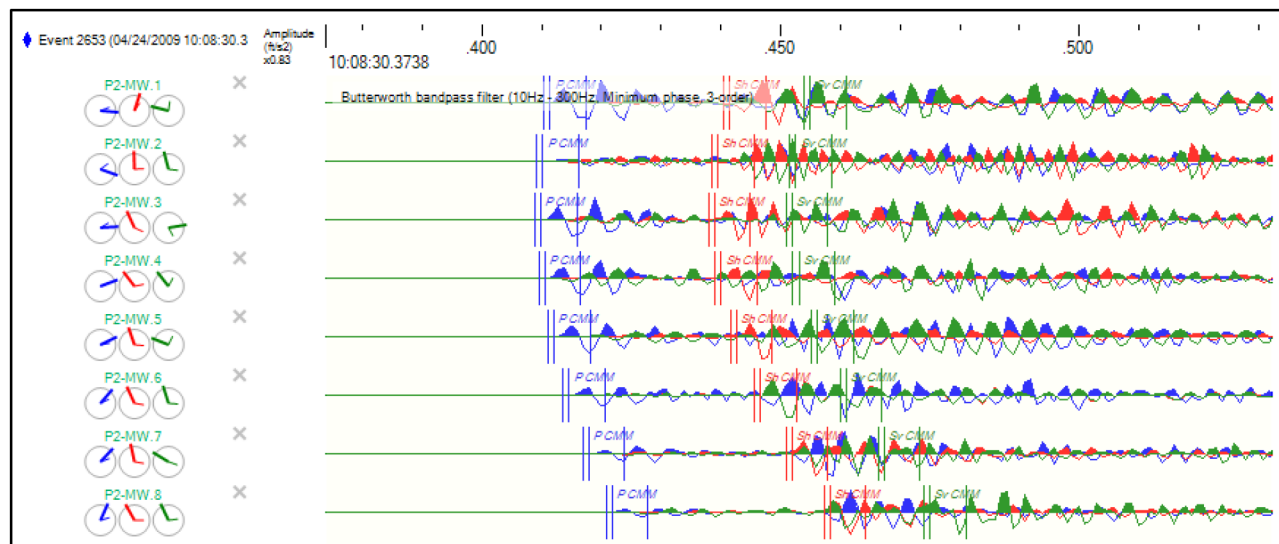


Figure 5—Three-Component Waveform Display for Event-1
Blue, red, and green traces represent P, Sh, and Sv components; colored ticks in circles on the left indicate hodogram angles for P, Sh, and Sv arrivals at the receiver

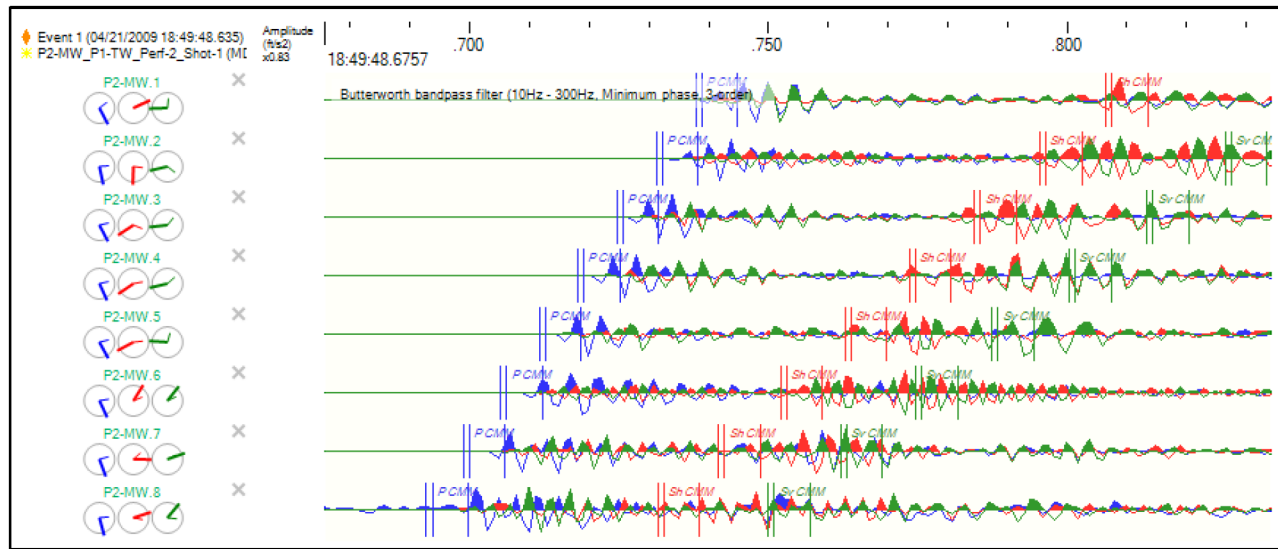


Figure 6—Three-Component Waveform Display for Event-2

The extracted source wavelet used in both forward and inverse modeling is shown in Figure 7. This wavelet, derived from Event-1, was validated for use across multiple stages of the workflow to ensure consistency and accurate time-domain modeling.

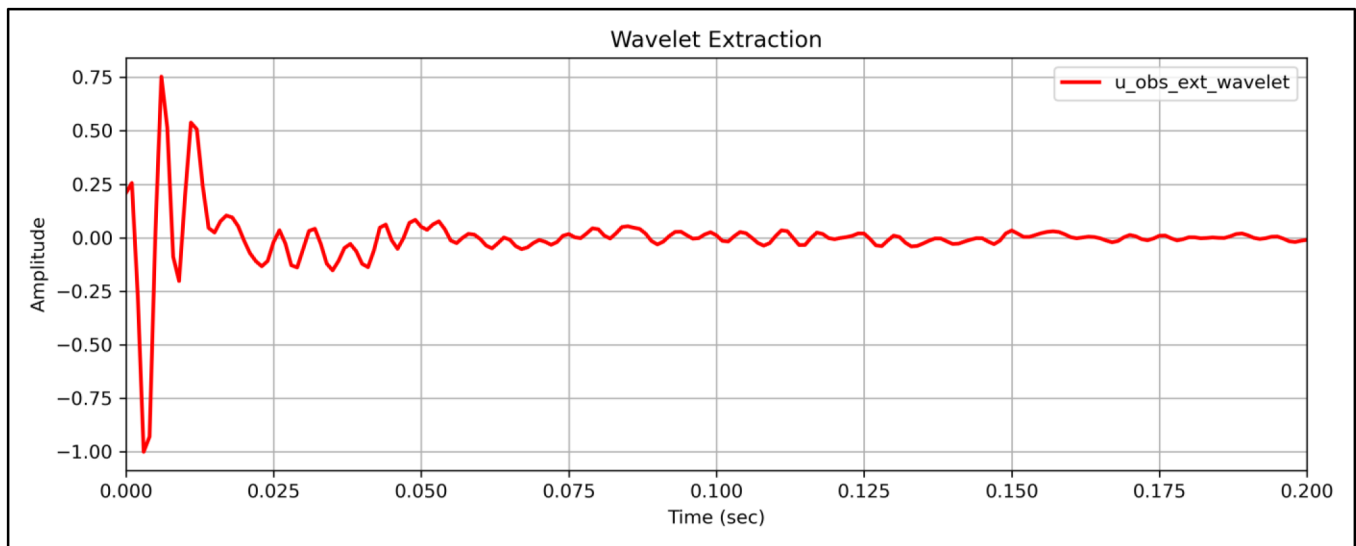


Figure 7—Extracted Source Wavelet from Event-1

Figure 8 compares the filtered observed waveforms (blue) with synthetic traces (red) generated using the inverted moment tensor from Event-1. The close match between the two confirms that the estimated source mechanism and model parameters are valid and that the inversion framework successfully reproduced the field-observed wavefield.

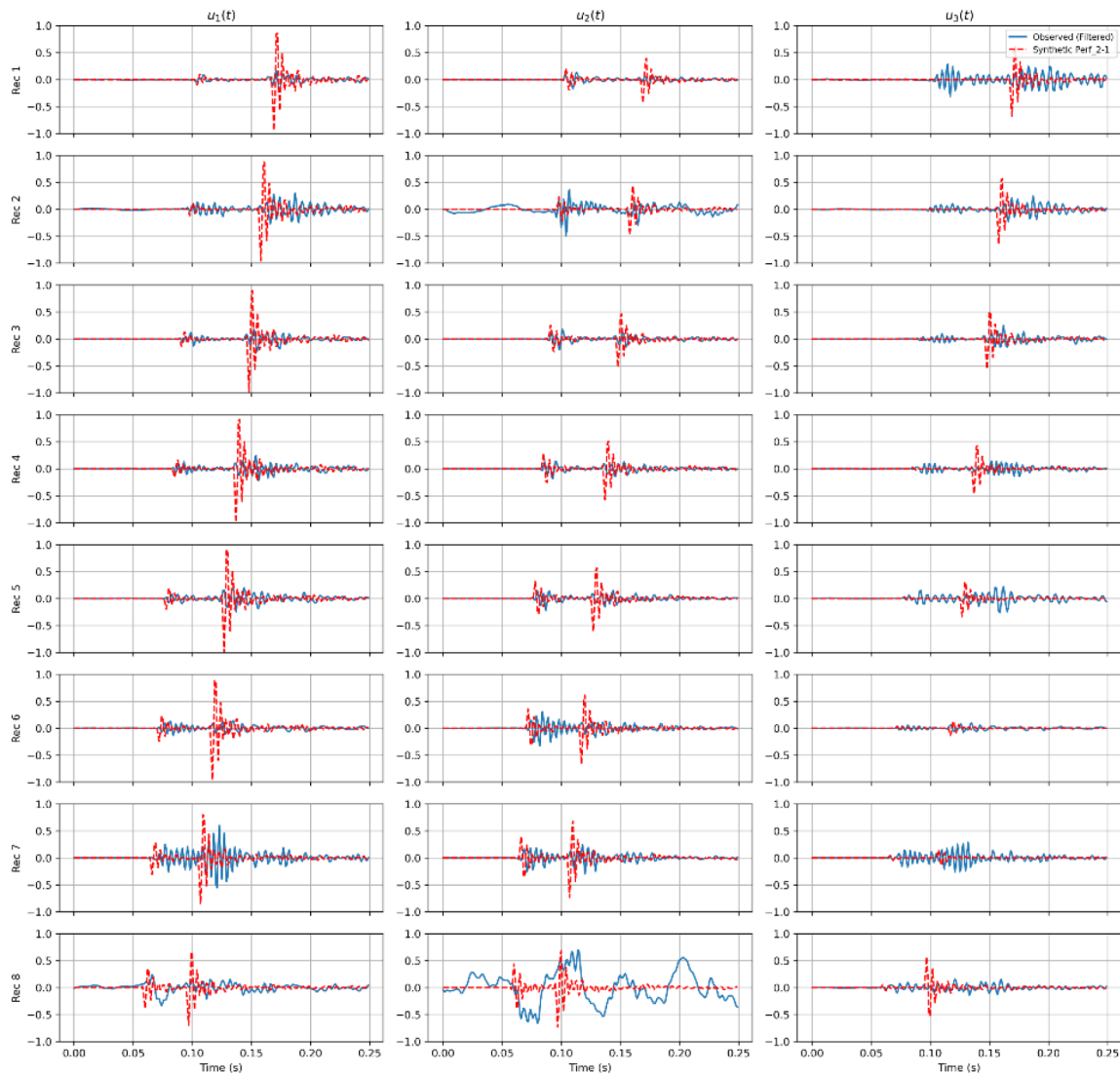


Figure 8—Observed vs. Synthetic Waveform Comparison – Event-1
 Overlay of filtered observed traces (blue) with synthetic traces (red) generated from the inverted MT solution. High waveform alignment demonstrates inversion accuracy.

Finally, [Figure 9](#) presents the full moment tensor inversion results for Event-1 and Event-2. Although both events were recorded using a horizontal borehole receiver array, the contrasting source-receiver geometries significantly impacted inversion resolution. Event-1 occurred near the center of the horizontal receiver array, benefiting from good azimuthal coverage and receiver distribution on both sides of the source. As a result, the inversion yielded a well-constrained moment tensor with strong resolution across most components, except for moderate uncertainty in the off-diagonal components M_{xz}/M_{zx} .

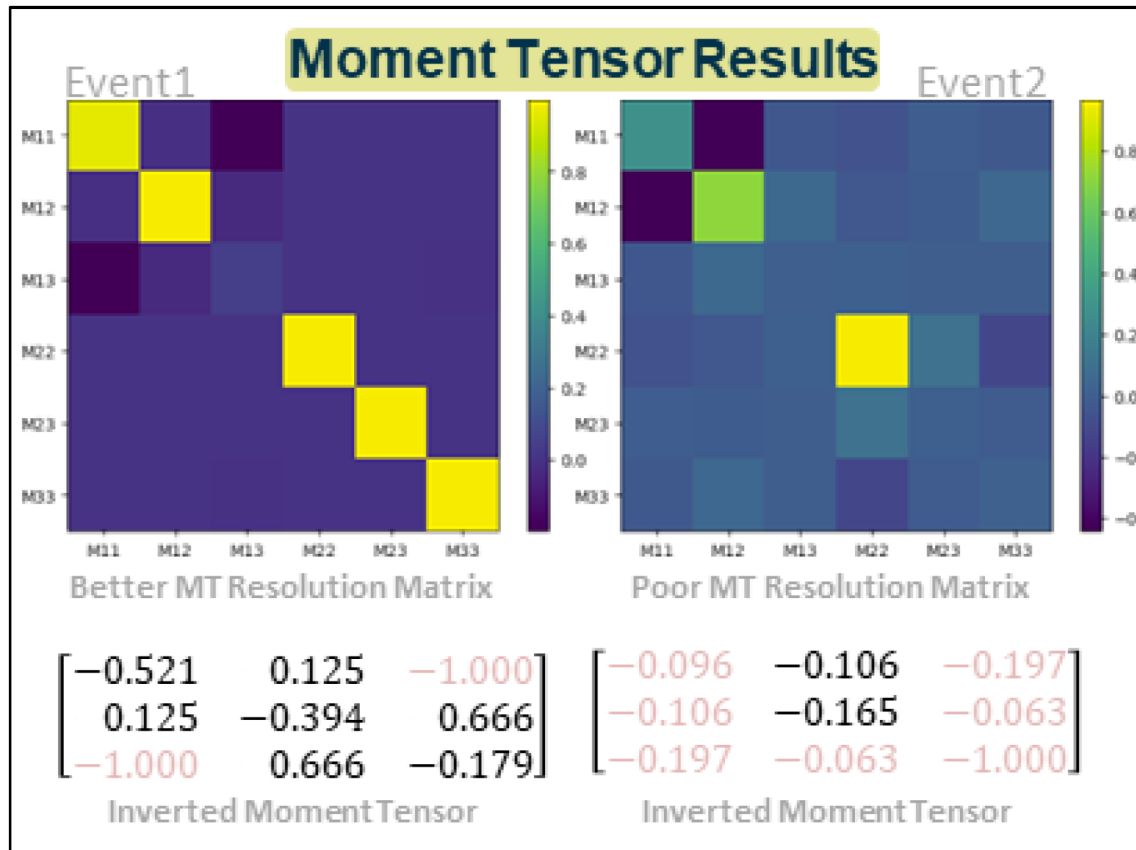


Figure 9—Field Moment Tensor Inversion Results – Event-1 vs. Event-2
 (Left) Event-1: Well-resolved moment tensor obtained from a microseismic event located near the center of a horizontal receiver array. All components except M_{xz}/M_{zx} show strong resolution.
 (Right) Event-2: Poorly resolved moment tensor due to limited azimuthal coverage; the event occurred south of the array with minimal aperture. Only M_{yy} is well constrained, while M_{xy}/M_{yx} show moderate reliability.
 Gray MT elements represent poorly constrained tensor elements.

In contrast, Event-2 was located almost entirely south of the receiver array, with limited azimuthal coverage and little to no receiver aperture surrounding the source. This unfavorable geometry led to poor resolution in most components. While M_{yy} was well resolved and M_{xy}/M_{yx} showed moderate reliability, the remaining components remained highly uncertain due to the lack of directional coverage. This comparison illustrates the critical importance of acquisition geometry—especially source-receiver azimuthal diversity—in determining the reliability of moment tensor inversion (Zhang, et al., 2016).

This detailed analysis across synthetic, and field datasets validates the eFWI-MTI workflow as a powerful tool for high-resolution microseismic source characterization. The ability to integrate both DAS and geophone recordings enhances the spatial density and diversity of input data, while the resolution matrix serves as a crucial diagnostic tool—both for inversion quality control and acquisition design optimization. Together, these capabilities enable a more comprehensive understanding of fracture processes, stress perturbations, and reservoir stimulation effectiveness.

CONCLUSIONS

This study demonstrates the implementation of an Elastic Full-Waveform Inversion (eFWI) framework integrated with Moment Tensor Inversion (MTI) for advanced hydraulic fracture monitoring. By directly using full waveform data in their native measurement domains—displacement for geophones and strain

for DAS—the framework overcomes critical limitations of conventional workflows, such as the need for manual time-picking and reliance on simplified kinematic models.

Synthetic tests confirmed the robustness of the inversion scheme, with high-resolution recovery of moment tensor components under both geophone and DAS-based acquisition setups. The DAS configuration, while directionally constrained, still achieved accurate inversion of principal components, validating its effectiveness when paired with well-planned survey geometry.

Field data analysis further demonstrated the method's adaptability to realistic acquisition scenarios. Event-1, occurring centrally within a horizontal receiver array, benefited from symmetric azimuthal coverage and resulted in a well-constrained moment tensor solution. In contrast, Event-2, positioned off-center with limited azimuthal support, exhibited poor resolution for most components. This contrast underscores the pivotal role of receiver geometry—particularly azimuthal aperture and bilateral sensor distribution—in successful MT inversion.

The resolution matrix emerged as a key diagnostic tool for evaluating inversion fidelity. It not only highlighted which moment tensor components were reliably recoverable but also revealed the limitations imposed by acquisition geometry. These insights are critical for both interpreting results and proactively designing optimal monitoring arrays.

In addition to strong waveform reconstruction and consistent inversion results, the integration of DAS and geophone recordings offered improved spatial density and observational diversity. This hybrid approach enhances sensitivity to a broader range of source mechanisms and strengthens the interpretation of fracture dynamics and stress redistribution.

Importantly, the ability to identify and characterize tensile events—those most strongly linked to productive fracture openings—positions this workflow as a valuable tool for real-time reservoir management and stimulation optimization.

However, despite the successful application in post-acquisition environments, real-time implementation of the eFWI-MTI framework remains computationally intensive. The demand for 3D elastic simulations, adjoint calculations, and iterative inversion places high pressure on compute resources. Ongoing work is focused on improving solver efficiency, leveraging parallel computing, and developing adaptive inversion strategies to facilitate near-real-time applications in the field.

In conclusion, the eFWI-MTI workflow delivers a high-resolution, fully automated, and physically informed solution for microseismic source characterization. Its applicability across sensing modalities and capacity to inform acquisition design and operational decisions mark it as a next-generation technology for intelligent hydraulic fracture monitoring.

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REFERENCES

- Aki, K. & Richard, P. G., 2002. *Quantitative Seismology*, Second Edition. s.l., s.n.
- Feng, F. et al, 2019. *A new method for evaluating the effectiveness of hydraulic fracturing in tight reservoirs*. s.l., Springer Verlag, pp. 1-12.
- Hartog, A. H., 2017. *An introduction to distributed optical fibre sensors*. s.l., CRC Press, pp. 1-440.

- Le Calvez, J., Hanson-Hedgecock, S. & Fredd, C. N., 2017. *Microseismic monitoring: A tool for evaluating fracture complexity in various geological settings*. s.l., s.n.
- Le Moigne, J. et al, 2014. *Georegistration of Earth Observing-1 (EO-1) data using Global Land Survey (GLS) maps*. s.l., Institute of Electrical and Electronics Engineers Inc., pp. 2518-2521.
- Li, J., Zhu, W., Biondi, E. & Zhan, Z., 2023. *Earthquake focal mechanisms with distributed acoustic sensing*. s.l., Nature Research, pp. 1-9.
- Li, L. et al, 2020. *Recent advances and challenges of waveform-based seismic location methods at multiple scales*. s.l., s.n.
- Li, Y., Karrenbach, M. & Ajo-Franklin, J. B., 2021. *Distributed Acoustic Sensing in Geophysics: Methods and Applications*. s.l., Wiley, pp. 1-301.
- Maxwell, S., 2014. *Microseismic Imaging of Hydraulic Fracturing*. s.l., Society of Exploration Geophysicists.
- Ma, Y., Eaton, D. W., Wang, C. & Aklilu, A., 2023. *Characterizing hydraulic fracture growth using distributed acoustic sensing- recorded microseismic reflections*. s.l., Society of Exploration Geophysicists, pp. WC47-WC57.
- Menke, W., 2012. *Geophysical Data Analysis: Discrete Inverse Theory*. s.l., Elsevier Inc..
- Pasyanos, M. E., Ichinose, G. A. & Ford, S. R., 2023. *Moment tensor event identification for collapses*. s.l., Oxford Academic, pp. 635-643.
- Pratt, R. G., 1999. *Seismic waveform inversion in the frequency domain, Part 1: Theory and verification in a physical scale model*. s.l., Soc of Exploration Geophysicists, pp. 888-901.
- Rashid, A. et al, 2025. *Research Advances on Distributed Acoustic Sensing Technology for Seismology*. s.l., Multidisciplinary Digital Publishing Institute, p. 196.
- Shekar, B. & Sethi, H. S., 2019. *Full-waveform inversion for microseismic events using sparsity constraints*. s.l., Society of Exploration Geophysicists, pp. KS1-KS12.
- Song, F. & Toksöz, M. N., 2011. *Full-waveform based complete moment tensor inversion and source parameter estimation from downhole microseismic data for hydrofracture monitoring*. s.l., Society of Exploration Geophysicists.
- Staněk, F., Jin, G. & Simmons, J., 2022. *Fracture Imaging Using DAS-Recorded Microseismic Events*. s.l., Frontiers Media S.A., p. 907749.
- Tape, W. & Tape, C., 2015. *A uniform parametrization of moment tensors*. s.l., Oxford Academic, pp. 2074-2081.
- Vavryčuk, V., 2015. *Inversion for the composite moment tensor*. s.l., Seismological Society of America, pp. 3024-3035.
- Vera Rodriguez, I. & Wuestefeld, A., 2020. *Strain microseismics: Radiation patterns, synthetics, and moment tensor resolvability with distributed acoustic sensing in isotropic media*. s.l., Society of Exploration Geophysicists, pp. KS101-KS114.
- Virieux, J. & Operto, S., 2009. *An overview of full-waveform inversion in exploration geophysics*. s.l., Society of Exploration Geophysicists.
- Wang, C. et al, 2022. *Hydraulic Fracture Monitoring Using Distributed Acoustic Sensing: A Case Study Using Dip-In Fiber*. s.l., OnePetro.
- Wang, H. & Alkhalifah, T., 2018. *Micro-seismic imaging using a source function independent full waveform inversion method*. s.l., Oxford University Press, pp. 46-57.
- Weaver, R. L. et al, 2008. Green's Function Reconstruction. *Seismic Interferometry*, pp. 99-329.
- Willacy, C. et al, 2019. *Full-waveform event location and moment tensor inversion for induced seismicity*. s.l., s.n.
- Yust, M. B. et al, 2023. *Near-Surface 2D Imaging via FWI of DAS Data: An Examination on the Impacts of FWI Starting Model*. s.l., Multidisciplinary Digital Publishing Institute, p. 63.
- Zhang, C. et al, 2019. *Automated microseismic event location by amplitude stacking and semblance*. s.l., Society of Exploration Geophysicists, pp. KS191-KS210.
- Zhang, H. et al, 2016. Discriminating induced seismicity from natural earthquakes using moment tensors and source spectra. *Journal of Geophysical Research: Solid Earth*, pp. 972-993.
- Zoback, M. D. & Harjes, H. P., 1997. Injection-induced earthquakes and crustal stress at 9 km depth at the KTB deep drilling site, Germany. s.l., American Geophysical Union, pp. 18477-18491.